

1.1/ Classify signals

a/ Closing price of utility stock

— one dimension, multichannel, discrete time, digital

Explain: + trading day $\rightarrow$ one variable $\rightarrow$ 1 dimension
+ Stocks $\rightarrow$ many company $\rightarrow$ multi channel
+ Closing price only $\rightarrow$ discrete-time
+ Stock price are in finite level $\rightarrow$ digital

1.3 / Periodic? Fundamental period?

a/ $x_a(t) = 3\cos(5t + \pi/6)$

Continuous time $\Rightarrow$ always periodic with $T = \frac{2\pi}{\omega} = \frac{2\pi}{5}$

b/ $x(n) = 3\cos(5n + \pi/6)$

$\omega = 2\pi f = 5 \Rightarrow f = \frac{5}{2\pi} \Rightarrow$ not rational $\Rightarrow$ non periodic

c/ $x(n) = 2\exp(j(n/6 - \pi))$

$\omega = 1/6 = 2\pi f \Rightarrow f = \frac{1}{12\pi} \Rightarrow$ not rational $\Rightarrow$ non periodic

d/ $x(n) = \cos(n/8) \cos(\pi n/8)$

$$= \frac{1}{2} \cos\left(\left(\frac{1}{8} - \frac{\pi}{8}\right)n\right) + \frac{1}{2} \cos\left(\left(\frac{1}{8} + \frac{\pi}{8}\right)n\right)$$

$f_1 = \frac{(1-\pi)}{16\pi}$, $f_2 = \frac{1+\pi}{16\pi} \Rightarrow$ not rational $\Rightarrow$ non periodic

e/ $x(n) = \cos\left(\frac{\pi n}{2}\right) - \sin\left(\frac{\pi n}{8}\right) + 3\cos\left(\frac{\pi n}{4} + \frac{\pi}{3}\right)$

$\omega_1 = 2\pi f_1 = \frac{\pi}{2} \Rightarrow f_1 = 1/4 \Rightarrow$ periodic, $N=4$

$\omega_2 = 2\pi f_2 = \frac{\pi}{8} \Rightarrow f_2 = 1/16 \Rightarrow$ periodic, $N=16$

$\omega_3 = 2\pi f_3 = \frac{\pi}{4} \Rightarrow f_3 = \frac{1}{8} \Rightarrow$ periodic, $N=8$

$\Rightarrow x(n)$ is periodic with $N=16$ (LCM of 4, 8, 16)

1.6 / $x_a(t)$, $T_p = \frac{1}{F_0}$ sampled at $F_s = \frac{1}{T}$ to produce $x(n) = x_a(nT)$

a/ Show that $x(n)$ is periodic if $\frac{T}{T_p} = \frac{k}{N}$

$$x(n) = A \cos(2\pi f_n t + \theta) \\ = A \cos\left(2\pi \frac{F_0 n}{F_s} + \theta\right)$$

$$= A \cos\left(2\pi \frac{T}{T_p} n + \theta\right), \quad \frac{T}{T_p} = f$$

$\Rightarrow x(n)$ is periodic $\Rightarrow \frac{T}{T_p} = \frac{k}{N}$ so it is rational

b/ $x(n)$ is periodic, what is T_p ?

$$f = \frac{k}{N} \Rightarrow T_d = \frac{k}{f} \cdot T = k \cdot \frac{T_p}{T} \cdot T = k \cdot T_p$$

c/ $T_d = k \cdot T_p \Rightarrow NT = k \cdot T_p \Rightarrow f = \frac{k}{N} = \frac{T}{T_p} \Rightarrow f$ is rational $\Rightarrow x(n)$ periodic

1.7 / Analog freq = 10 kHz

a/ Range of freq for reconstruction?

Nyquist theorem

$$F_s \geq 2F_{\max} \Rightarrow F_s \geq 20 \text{ kHz}$$

b/ $F_s = 8 \text{ kHz}$, $F_1 = 5 \text{ kHz}$

$$F_{\text{fold}} = \frac{8 \text{ kHz}}{2} = 4 \text{ kHz}$$

$F_{\text{fold}} < F_1 \Rightarrow$ Aliasing

$$F_{\text{aliasing}} = |5 - 8| = 3 \text{ kHz}$$

c/ $F_s = 8 \text{ kHz}$, $F_2 = 9 \text{ kHz}$

$$F_{\text{fold}} = 4 \text{ kHz}, F_{\text{fold}} < F_2$$

$$F_{\text{alias}} = |9 - 8| = 1 \text{ kHz}$$

1.9 / $x_a(t) = \sin(480\pi t) + 3\sin(720\pi t)$, $F_s = 600$

a/ Nyquist Sampling rate?

$$\omega_1 = 480\pi = 2\pi f_1 \Rightarrow f_1 = 240 \text{ Hz}$$

$$\omega_2 = 720\pi = 2\pi f_2 \Rightarrow f_2 = 360 \text{ Hz}$$

$$F_{\max} = 360 \quad | \quad F_n = 2F_{\max} = 720 \text{ Hz}$$

b/ F_{fold} ?

$$F_{\text{fold}} = F_s/2 = 300 \text{ Hz}$$

c/ f ?

$$\begin{aligned} x(n) &= x_a(Nt) \\ &= x_a\left(\frac{n}{F_s}\right) = \sin\left(\frac{480\pi n}{600}\right) + 3\sin\left(\frac{720\pi n}{600}\right) \\ &= \sin\left(\frac{4\pi n}{5}\right) - 3\sin\left(\frac{4\pi n}{5}\right) \\ &= -2\sin\left(\frac{4\pi n}{5}\right) \end{aligned}$$

$$\Rightarrow \omega = \frac{4\pi}{5}$$

d/ $x(n)$ pass ideal D/A, reconstructed $y_a(t)$?

$$y_a(t) = x(F_s t) = -2\sin(480\pi t)$$

$$\left(F = \frac{4/5\pi}{2\pi} \times 600 = 240 \text{ Hz} \right)$$

1.10 / $x_a(t) = 3\cos(600\pi t) + 2\cos(1800\pi t)$
10000 bps, 1024 levels

a/ Sampling freq? Folding freq?

$$\cdot \log_2 1024 = 10$$

$$F_s = \frac{10000}{10} = 1000 \Rightarrow F_s = 1000 \text{ Hz}$$

$$F_{\text{folding}} = \frac{1000}{2} = 500 \text{ Hz}$$

b/ Nyquist?

$$F_{\max} = \frac{1800\pi}{2\pi} = 900 \text{ Hz}$$

$$F_{\text{Nyquist}} = 2 F_{\max} = 1800 \text{ Hz}$$

c/ f?

$$f_1 = \frac{600\pi}{1000} = 0,6 \pi$$

$$f_2 = \frac{1800\pi}{1000} = 1,8 \pi = +0,2 \pi$$

d/ Δ ?

$$\Delta = \frac{5 - (-5)}{L - 1} = \frac{10}{1023}$$